

Reproducibility Checklist

This checklist is filled for the current BGR submission package. The main paper is anonymous; commands, configs, per-seed CSVs, and run logs are listed in the repository's `README.md` and `results/README.md`.

1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) **yes**. The method section defines replayable states, recovery curves, critical radii, BGR priority, and includes a training-loop figure.
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) **yes**. The paper marks OpenVLA recovery-curve analysis as fixed-policy evidence, BGR-Suffix fine-tuning as future work, and tabular grid-policy failures as negative diagnostics.
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) **yes**. The related-work section cites curriculum learning, PLR, PAIRED/ACCEL, domain randomization, MiniGrid, LIBERO, and OpenVLA.

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) **no**. The paper introduces definitions and an algorithmic replay principle, but does not claim new theorems.

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) **NA**.
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) **NA**.
- 2.4. Proofs of all novel claims are included (yes/partial/no) **NA**.
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) **NA**.
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) **NA**.
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) **NA**.
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) **NA**.

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) **partial**. Main BGR training results use procedural synthetic/grid/suffix generators. The LIBERO/OpenVLA recovery audit uses existing benchmark task definitions, init states, and fixed-policy rollout artifacts to validate learned-policy recovery-curve measurement.

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) **yes**. The paper motivates synthetic recovery curves, procedural grid-margin replay states, robot suffix diagnostics, and LIBERO/OpenVLA as a realistic manipulation-policy target.
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) **yes**. The experiments are generated by checked-in scripts/configs; per-seed result CSVs are included under `results/`.
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **yes**. The generated procedural data and code are included in the repository under an MIT license.
- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) **yes**. LIBERO and OpenVLA are cited; MiniGrid is cited for grid-style context.
- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) **yes** for LIBERO task definitions, benchmark assets, and OpenVLA model code/weights. The checked-in repository contains compact derived recovery summaries; raw fixed-policy rollout artifacts are referenced by cluster path.
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) **NA**. No non-public dataset is used for claimed quantitative results.

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) **yes**.

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) **par-**

tial. Final configs and ablation variants are checked in; the full search history is not exhaustively enumerated in the main paper.

- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) [yes](#). [Procedural generation, LIBERO probing, OpenVLA recovery summarization, aggregation, and figure scripts are included.](#)
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) [yes](#). [Experiment scripts, configs, tests, aggregation, and result CSVs are checked in.](#)
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) [yes](#). [The repository is on GitHub and includes an MIT license.](#)
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) [partial](#). [Code is modular and tested; comments are intentionally sparse and do not yet cross-reference paper sections.](#)
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) [yes](#). [Every config lists seeds; scripts pass seeds to NumPy generators and result CSVs include per-seed rows.](#)
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) [yes](#). [The run ledger records Slurm partitions, CPU/memory/time requests, hostnames for logged runs, environment constraints, and JSON environment snapshots for compute and GPU jobs.](#)
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) [yes](#). [Recovery AUC, critical radius, clean success, transfer RAUC, and AULC are defined or described.](#)
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) [yes](#). [Tables and text state three- or five-seed means as applicable.](#)
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no)

[yes](#). [Aggregated tables include standard errors; experiments report median critical radius and AULC in addition to final means.](#)

- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) [yes](#). [The repository includes paired exact sign-flip tests for the reported seed-paired comparisons. Because the current diagnostic runs use only three or five seeds, the paper interprets these tests descriptively and does not overstate thresholded significance.](#)
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) [yes](#). [Final hyperparameters are in the checked-in YAML configs under `configs/`.](#)